Preview: set-01/by_level/Grade_03-04_Ecolier/y2024_q21 (v2)

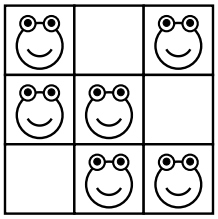
Two jumps that keep every row and column count: the second jump must undo the first

Part A: Each question is worth 3 points

1. There are exactly 2 frogs in each row and in each column of this 3 × 3 pond. A frog can jump to a **neighbouring** field (left, right, up or down) if it is empty.

Only **one** frog jumps.

How many different jumps are possible? (Count every frog and every empty field it can reach.)

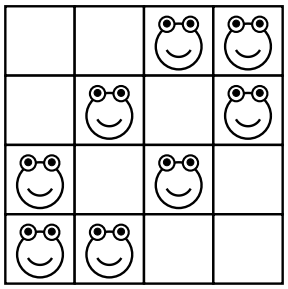


A. 3 ***B.*** 6 ***C.*** 8 ***D.*** 10 ***E.*** 12

2. There are exactly 2 frogs in each row and in each column of this 4 × 4 pond. A frog can jump to a **neighbouring** field (left, right, up or down) if it is empty.

Only **one** frog jumps.

How many different jumps are possible? (Count every frog and every empty field it can reach.)



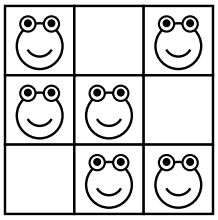
A. 8 ***B.*** 14 ***C.*** 16 ***D.*** 17 ***E.*** 18

Part B: Each question is worth 4 points

3. There are exactly 2 frogs in each row and in each column of this 3 × 3 pond. A frog can jump to a **neighbouring** field (left, right, up or down) if it is empty.

Two of the frogs jump at the same time, each to an empty neighbouring field. After that there are again exactly 2 frogs in each row and in each column.

How many possibilities are there for two frogs to jump like this?

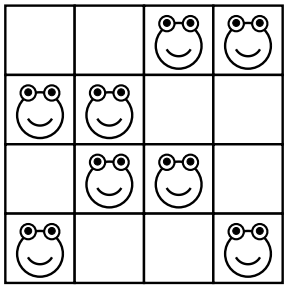


A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

4. There are exactly 2 frogs in each row and in each column of this 4 × 4 pond. A frog can jump to a **neighbouring** field (left, right, up or down) if it is empty.

Two of the frogs jump at the same time, each to an empty neighbouring field. After that there are again exactly 2 frogs in each row and in each column.

How many possibilities are there for two frogs to jump like this?

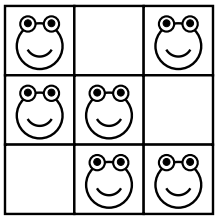


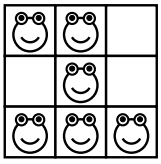
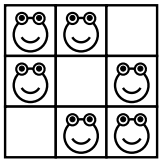
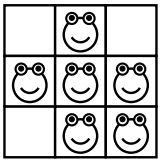
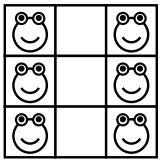
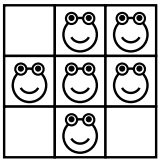
A. 12 ***B.*** 13 ***C.*** 14 ***D.*** 15 ***E.*** 24

5. There are exactly 2 frogs in each row and in each column of this 3 × 3 pond. A frog can jump to a **neighbouring** field (left, right, up or down) if it is empty.

Two frogs jump at the same time, each to an empty neighbouring field. After that there are again 2 frogs in each row and in each column.

Which picture can show the pond **after** the jump?

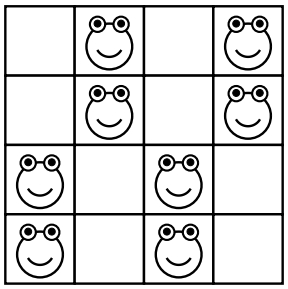


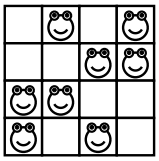
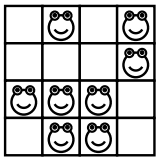
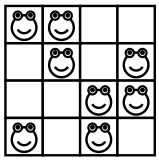
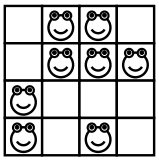
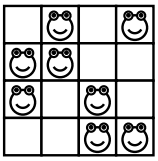
A.  ***B.***  ***C.***  ***D.***  ***E.*** 

6. There are exactly 2 frogs in each row and in each column of this 4 × 4 pond. A frog can jump to a **neighbouring** field (left, right, up or down) if it is empty.

Two frogs jump at the same time, each to an empty neighbouring field. After that there are again 2 frogs in each row and in each column.

Which picture can show the pond **after** the jump?



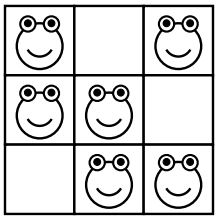
A.  ***B.***  ***C.***  ***D.***  ***E.*** 

Part C: Each question is worth 5 points

7. There are exactly 2 frogs in each row and in each column of this 3 × 3 pond. A frog can jump to a **neighbouring** field (left, right, up or down) if it is empty.

This time only **one** frog jumps, to an empty neighbouring field. After the jump there must still be exactly 2 frogs in each row and in each column.

How many different jumps are possible?

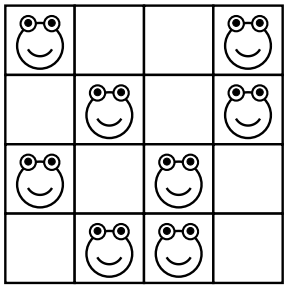


A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 4 ***E.*** 8

8. There are exactly 2 frogs in each row and in each column of this 4 × 4 pond. A frog can jump to a **neighbouring** field (left, right, up or down) if it is empty.

This time only **one** frog jumps, to an empty neighbouring field. After the jump there must still be exactly 2 frogs in each row and in each column.

How many different jumps are possible?

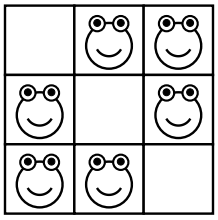


A. 0 ***B.*** 2 ***C.*** 4 ***D.*** 15 ***E.*** 18

9. There are exactly 2 frogs in each row and in each column of this 3 × 3 pond. A frog can jump to a **neighbouring** field (left, right, up or down) if it is empty.

Two frogs jump at the same time, each to an empty neighbouring field, and after that there are again 2 frogs in each row and in each column.

How many of the frogs can **never** be one of the two jumping frogs?

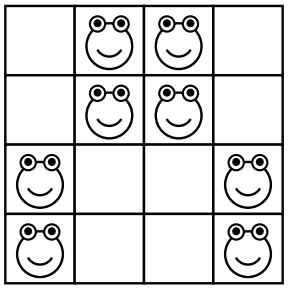


A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 6

10. There are exactly 2 frogs in each row and in each column of this 4 × 4 pond. A frog can jump to a **neighbouring** field (left, right, up or down) if it is empty.

Two frogs jump at the same time, each to an empty neighbouring field, and after that there are again 2 frogs in each row and in each column.

How many of the frogs can **never** be one of the two jumping frogs?



A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 8

Answer Key

1. **C**	**2. C**	**3. D**	**4. A**	**5. B**	**6. A**
7. **A**	**8. A**	**9. C**	**10. A**		

Solutions

1. C (8) — Go frog by frog and count its empty neighbours: 1 + 2 + 1 + 2 + 1 + 1 = 8.
[set-01/by_level/Grade_03-04_Ecolier/2024 Q21 · one_frog]

2. C (16) — Go frog by frog and count its empty neighbours: 2 + 0 + 4 + 2 + 2 + 4 + 0 + 2 = 16.
[set-01/by_level/Grade_03-04_Ecolier/2024 Q21 · one_frog]

3. D (4) — If a frog jumps sideways, one column loses a frog and the next one gets one more - the other frog must jump sideways between the same two columns, the other way, in another row (the same for jumps up and down between two rows). Checking each pair of neighbouring columns and rows gives 4.
[set-01/by_level/Grade_03-04_Ecolier/2024 Q21 · count_pairs]

4. A (12) — If a frog jumps sideways, one column loses a frog and the next one gets one more - the other frog must jump sideways between the same two columns, the other way, in another row (the same for jumps up and down between two rows). Checking each pair of neighbouring columns and rows gives 12.
[set-01/by_level/Grade_03-04_Ecolier/2024 Q21 · count_pairs]

5. B — First count the frogs in every row and column of each picture. Then look at which frogs moved: in the right picture each of the two frogs moved just one field (not diagonally), and the second jump undoes the first.
[set-01/by_level/Grade_03-04_Ecolier/2024 Q21 · which_picture]

6. A — First count the frogs in every row and column of each picture. Then look at which frogs moved: in the right picture each of the two frogs moved just one field (not diagonally), and the second jump undoes the first.
[set-01/by_level/Grade_03-04_Ecolier/2024 Q21 · which_picture]

7. A (0) — None! A sideways jump takes a frog out of one column and puts it into another, so one column has 1 and one has 3. A jump up or down does the same to two rows. The answer is 0. [set-01/by_level/Grade_03-04_Ecolier/2024 Q21 · one_frog_trick]

8. A (0) — None! A sideways jump takes a frog out of one column and puts it into another, so one column has 1 and one has 3. A jump up or down does the same to two rows. The answer is 0. [set-01/by_level/Grade_03-04_Ecolier/2024 Q21 · one_frog_trick]

9. C (2) — Find all the ways (4): the second jump must undo the first one (sideways between the same two columns, or up/down between the same two rows). 4 frogs take part in some way, so 6 - 4 = 2 never jump.
[set-01/by_level/Grade_03-04_Ecolier/2024 Q21 · never_jumps]

10. A (0) — Find all the ways (12): the second jump must undo the first one (sideways between the same two columns, or up/down between the same two rows). 8 frogs take part in some way, so 8 - 8 = 0 never jump.
[set-01/by_level/Grade_03-04_Ecolier/2024 Q21 · never_jumps]